Wu-Xing dynamics
Review and writing plan

For the existing manuscript, Sections 3 and 4 | Revised 4 October 2026

Assessment

The core equilibrium, Jacobian, baseline eigenvalue and perturbation calculations are sound. They provide enough material for a focused results write-up now. Several conclusions in the Claude record go beyond what the computations establish; the regime classification and some bifurcation claims need substantive correction.

Recommended scope: summarize equilibrium stability, responses to disturbances, parameter maps and sampling sensitivity. Use the nonnegative five-term modified model throughout Section 4. Preserve the mentor's actual section titles: *Dynamics of Lin's Model* and *A modified version of the model and its dynamics*. The older SURE-template instructions in the record are historical context, not the structure requested here.

Independently verified baseline results

Quantity	Lin, (a,b,c,d,g,h)=(1,1,1,1,1,1)	Simple modified model, (a,k,c,g,h)=(2,.5,1,.5,.3)
Balanced value	1/3 = 0.333333	(3 + sqrt(59))/4 = 2.670286
Largest eigenvalue real part	+0.206011	-1.153724
Local conclusion	Unstable equilibrium	Locally exponentially stable equilibrium
Wood +5%, final distance at t=120	0.8776	About 1e-11 or smaller
Wood reduced to 30%, final distance at t=120	0.9899	About 1e-11 or smaller

Distances are Euclidean distances to the relevant balanced vector. The two baselines use different parameters and activity scales. They demonstrate behavior at specified settings; they do not establish a calibrated comparison of physiological recovery speeds.

A defensible central claim

At the all-ones parameter setting, Lin's positive balanced equilibrium is unstable. The simple modified model has a locally stable positive balanced equilibrium at its chosen baseline and recovers from the tested disturbances. Under the specified parameter sweeps and sampling distributions, the modified model more frequently admits a locally stable positive balanced equilibrium.

Do not claim: Lin can never have stable balance; the modified model is globally stable; saturation alone causes the difference; or the experiments validate health, disease or treatment predictions.

What this review actually checked

The equations and figure-generating code in the supplied record were reviewed and the main results independently recomputed. The record's linked *figures/* folder was not supplied. Therefore this is a mathematical and computational audit with regenerated figures, not a pixel-by-pixel certification of every historical image. The original Lin paper was also checked against the local PDF.

Which figures are ready to use?

Figure names below refer to Part 9 of the supplied record unless another part is named. Older notebooks repeat many of the same experiments.

Figure family	Verdict	Action before manuscript use
Eigenvalue plane (master Fig. 1)	Verified	Use. State both full parameter tuples and label this local stability of the balanced equilibrium.
Small/severe perturbations (Fig. 2)	Verified	Use. Call the large disturbance a tested finite perturbation. It is not a global-stability proof. Show the early recovery interval for the modified model.
Five organs in turn (Fig. 3)	Verified idea; redundant	Cyclic symmetry makes these equivalent tests up to relabeling. Keep as a numerical consistency check or supplement.
Stability heatmaps (Fig. 4)	Verified values; relabel	Use the master mapping: received restraint is d in Lin, h in the modified model. Explicitly distinguish missing positive equilibria and the percentage denominator.
Regime map and regime shares (Figs. 5, 7)	Not reliable as dynamical classifications	Do not publish chaos percentages or multistability claims from this classifier. Rebuild with unresolved, numerical failure and threshold-exceeded categories.
Monte Carlo stability bars (Fig. 6)	Verified numbers; conditional interpretation	Report the sampling distribution, seed, sample size and valid counts. Give stability among valid draws and among all draws. Priors are modeling assumptions.
CHILD-term ablation (Fig. 8)	Verified within its experiment	Optional sensitivity result. It does not isolate the functional-form change or establish a uniquely fair comparison.
Example trajectories (Figs. 9, 10)	Consistent; largely duplicate Fig. 2	Use only if a time-series/phase-portrait pair communicates something additional. Avoid magnifying numerical noise around the stable point.
P4 mode map / Hopf claim	Formula valid; figure and claim need repair	At c=1 the crossing is beyond the plotted h range. The c=2 reported endpoint is not a crossing. A nondegenerate Hopf assertion needs more than a nonzero imaginary part.
P5 bistability example	Verified for its chosen parameters	Optional supplement. The threshold is for synchronous starts; the basin boundary in five dimensions is not a scalar threshold.

Recommended main set: eigenvalue comparison, perturbation recovery, corrected stability maps, and one concise sampling table or plot. This is enough for the mentor's immediate request. Keep the rest as supporting material or future work.

Verified baseline figures

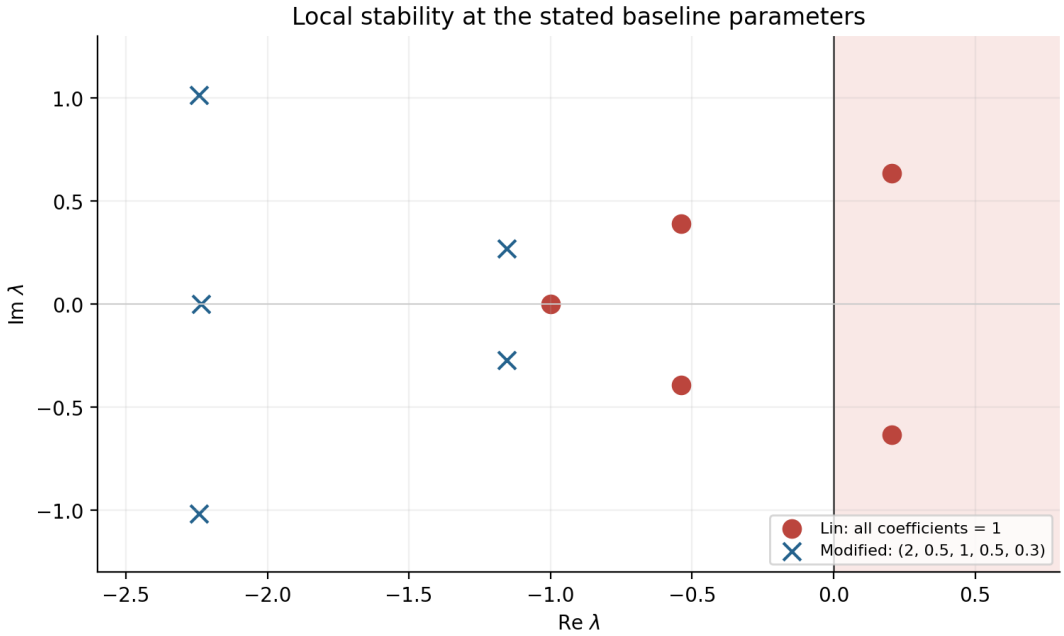


Figure 1. Jacobian eigenvalues at the positive balanced equilibrium for the two stated baselines. Lin has a complex pair with positive real part. All five eigenvalues of the simple modified model have negative real parts. The magnitudes are in each model's stated time units.

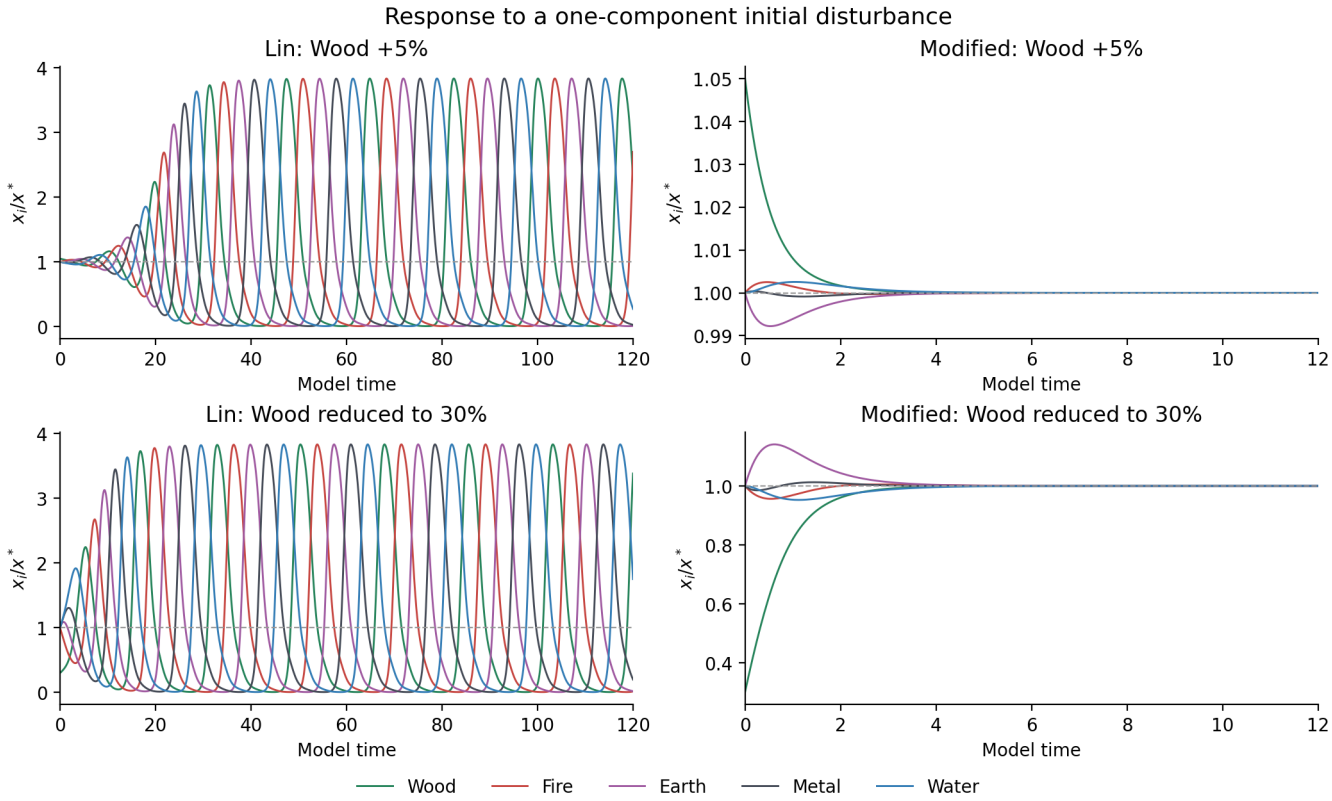


Figure 2. Wood is initially raised by 5% (top) or reduced to 30% of its equilibrium value (bottom); other components start at balance. Vertical axes show activity divided by the relevant equilibrium value. Lin is displayed to t=120; the modified model is displayed to t=12 to make recovery visible. All runs were integrated to t=120.

Corrected maps and sampling presentation

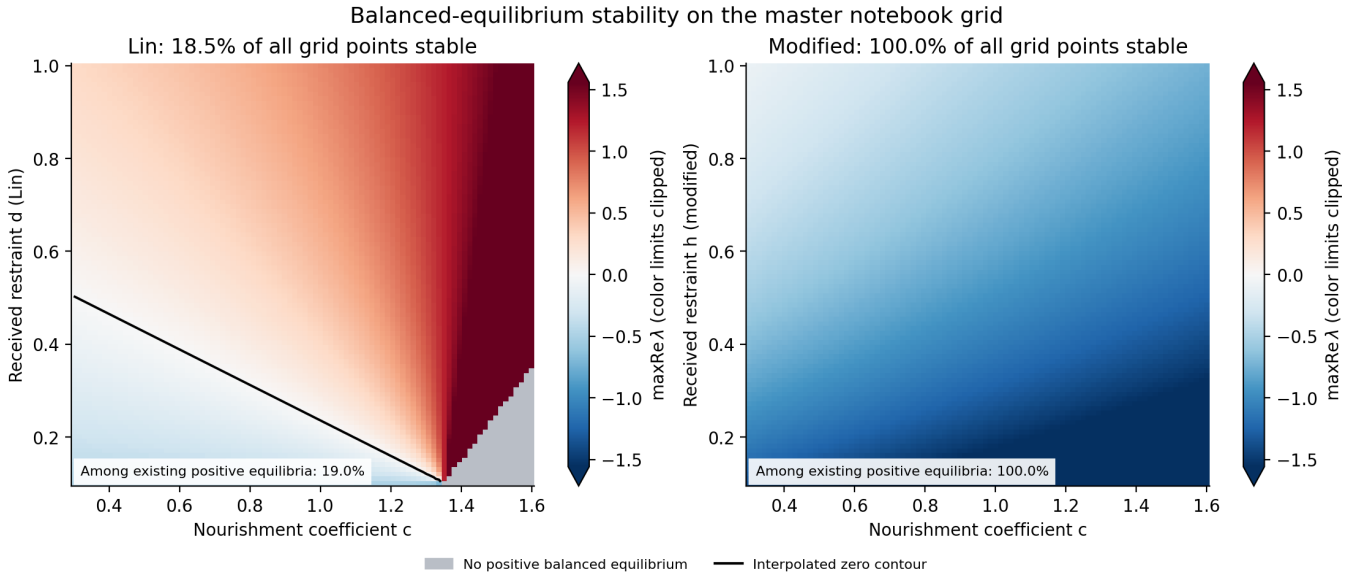


Figure 3. A 90 x 90 grid over c in [0.3,1.6] and received restraint in [0.1,1]. Fixed values: Lin a=1.5, b=0.65, g=h=0.3; modified a=1.5, k=0.65, g=0.3. Gray denotes no positive balanced equilibrium. The black line is a numerical zero contour. Color limits are clipped at the pooled 90th percentile of absolute spectral values; endpoints on the color bars show clipping.

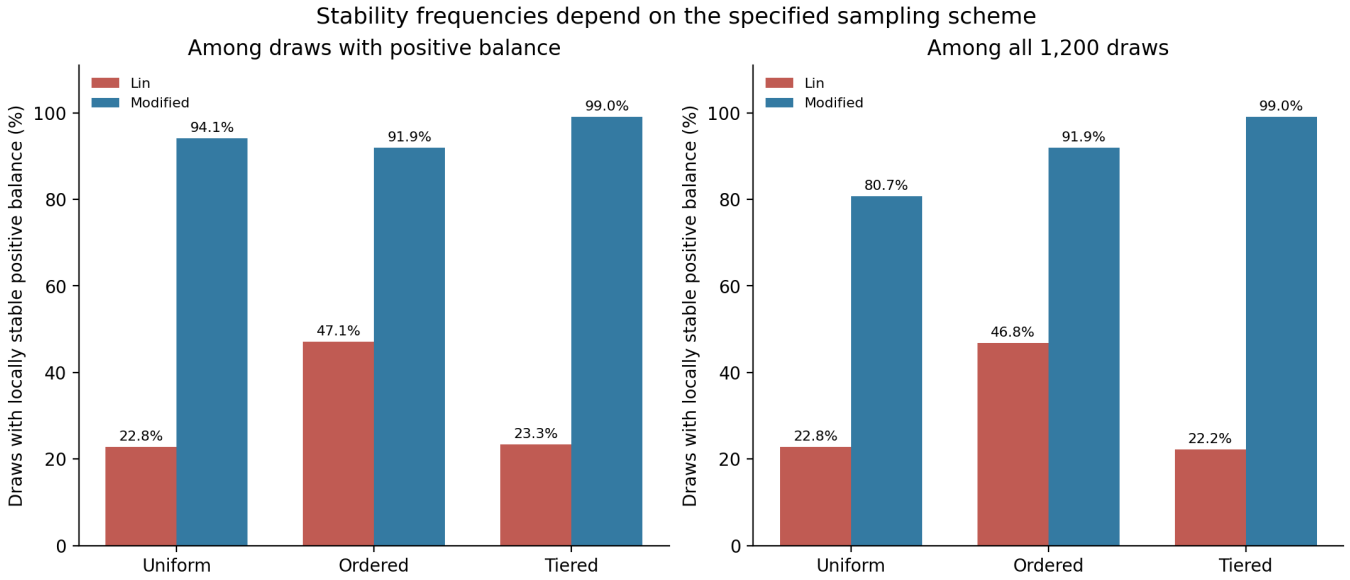


Figure 4. Identical master-notebook draws and seed (1,200 draws per scheme, seed 3). Left: the fraction stable among draws with a positive balanced equilibrium. Right: the fraction of all draws that admit such a stable equilibrium. These describe the specified distributions, not a probability that a model is correct or physiologically healthy.

Corrections needed in the mathematical narrative

1. Lin can have stable positive balance

For positive a and M=b+d+g+h-c, the synchronous value is x*=a/M. At that equilibrium, with theta_j=2*pi*j/5, the eigenvalue real parts are:

```
Re(lambda_j) = x*[-b + (c-g)cos(theta_j)
- (d+h)cos(2 theta_j)].
```

This reproduces the baseline eigenvalues. The concrete counterexample (a,b,c,d,g,h)=(1,2,1,1,1,1) has x*=0.25 and largest real part -0.095492. Thus the statement that Lin cannot represent stable balance is false. Use it as a one-sentence qualification, without turning the section into a new theorem paper.

2. Local stability, boundedness and equilibrium existence differ

Strictly negative eigenvalue real parts imply local exponential stability. A positive real part implies instability. A zero real part requires further analysis. Bounded trajectories need not approach an equilibrium. Convergence from several starts is not global stability, and failure to find a positive synchronous equilibrium does not make an ODE ill posed.

For Lin on the synchronous subspace, s'=a*s-M*s^2. With a>0 and positive s(0), M<0 gives finite-time blow-up; M=0 gives exponential growth, not finite-time blow-up; M>0 gives convergence to a/M on this subspace. Do not extend the last conclusion to every nonsynchronous trajectory without a separate argument.

The record's c=3.9 symmetric stress test reports a maximum near 131 from s(0)=1, despite M=0.1. The exact scalar trajectory increases monotonically to 10. That excursion cannot be a feature of the exact symmetric solution; numerical loss of symmetry must be investigated before using it as evidence.

3. The simple modified model has a valid boundedness argument

Assume nonnegative coefficients and k>0. At x_i=0 its derivative is c*x_(i-1), which is nonnegative. At a component attaining the maximum X, the derivative is at most a*X-k*X^2+c*X/(1+X), and hence at most a*X-k*X^2+c. Therefore the box [0,B]^5 is positively invariant and attracts from above, where:

```
B = [a + sqrt(a^2 + 4*k*c)]/(2*k).
```

For arbitrary nonnegative initial data, max_i x_i(t) is bounded by max{max_i x_i(0),B}. This is a boundedness statement for the simple model, not a convergence proof.

4. Describe what actually changed between the models

The simple modified model omits the direct CHILD loss, changes the nourishment law, and changes the target dependence. In Lin, the target derivative is -h*x_i; in the modified model it is +g*x_i/(1+x_target)^2. The target interaction therefore changes sign. The nourishment c*x_mother/(1+x_i) is linear in the mother and decreases with recipient activity; it is not a conventional saturation in the mother variable. Avoid saying the models have identical interactions differing only by saturation.

5. Parameter windows are assumed scenarios

The proposed ordering and tiered ranges are not fitted or physiologically established. Raw coefficients multiplying different functional forms are not automatically comparable in strength; scaling, units and activity level matter. Describe these as sensitivity scenarios, and report them explicitly.

Regime classification and bifurcation figures

Regime classification: hold these figures back

The classifier labels fewer than four peaks as periodic; labels irregular peaks as chaotic; and treats solver failures or crossing an activity cap as blow-up. None is a sufficient diagnosis. A drifting transient may produce few peaks, a quasiperiodic orbit may look irregular, and a bounded orbit may exceed a chosen cap. A three-way tie is resolved by sorted-label order, so the plotted result is not always a majority.

The all-equal start (0.5,...,0.5) is also special: exact trajectories remain synchronous. At Lin's baseline the uniform eigenvalue is -1 while the transverse leading real part is +0.206011. Such a trajectory can converge to an equilibrium that repels generic perturbations. Disagreement with nonsynchronous starts does not establish coexisting attractors. Near-synchronous randomized starts and convergence checks are needed for a basin study.

The conservative Lin case does not have established chaos here

Independent variational/QR integration confirms shrinking finite-time leading estimates: T=500: 0.01289, T=1000: 0.00729, T=1500: 0.00487, T=2000: 0.00387. These results do not reproduce the published exponent near 0.67. Keep the conclusion at non-reproduction; neither irregular pictures nor finite-time near-zero exponents prove chaos or exclude every form of weak chaos. The all-ones orbit has a reproduced period near 17.626 and a spectrum consistent with an attracting periodic orbit.

P4 and P5: retain qualified statements only

At (a,k,c,g)=(1.5,0.65,1,0.5), the true crossing is h=2.371813, with imaginary part about 0.310613 and nonzero crossing speed. P4's right-hand slice ends at h=2, so it does not show that crossing. The record's c=2, h=4 row has negative leading real part and must not be called a Hopf point. A sign-checked search finds the first crossing at h=4.517499. A nondegenerate Hopf claim still requires the relevant nonlinear nondegeneracy check.

P5 is valid for its chosen parameters a=2,k=0.5,c=0.1,h=0.3: the synchronous fold is at g=2.55 and x*=0.75. At g=2.5 the origin and upper equilibrium x*=1 are locally stable; x*=0.5 is unstable. The 0.5 threshold shown applies to all-equal initial states. For general parameters, two positive roots require a>k+h and do not by themselves ensure transverse stability of the upper branch. The generic transcritical description also needs a different from k+h. At the exact fold, classify the repeated root as nonhyperbolic, not separately stable and unstable.

Plan for Section 3: Dynamics of Lin's Model

Purpose: establish what the original model does and what the existing numerical analysis adds. Aim for a concise sequence from equation to result to limitation. This can be drafted now without completing the later theoretical program.

3.1 Model and the positive balanced equilibrium

Introduce the five variables in generating order and define indices modulo five. Write Lin's six-term equation, assume symmetric nonnegative coefficients, and identify the mother, child, controller and target terms. State that the current comparison uses no external forcing. Define the positive balanced equilibrium x*1, with x*=a/(b+d+g+h-c) when a>0 and the denominator is positive.

Then give the stability procedure once for both sections: evaluate the Jacobian at that equilibrium and compute alpha=max Re(lambda). Explain negative, positive and zero values. Specify that this concerns the balanced equilibrium, not all possible equilibria or attractors. A full Jacobian may go in an appendix; the Fourier formula is optional supporting mathematics.

3.2 Baseline instability and response to disturbances

State the all-ones parameter tuple before presenting numbers. Give x*=1/3 and the destabilizing pair 0.206011 +/- 0.634038i. Reference the Lin part of the eigenvalue figure. Then describe the initial-disturbance protocol: start at balance, change only Wood by +5% or multiply it by 0.3, and integrate to t=120.

Report departure from balance and the subsequent oscillatory behavior observed in the trajectory. Distinguish the unstable equilibrium from an attracting periodic orbit. If the Lin reproduction is included, give the separate start (1,2,3,4,5) and period near 17.626. Avoid saying that the severe perturbation causes permanent organ extinction; the tested baseline trajectories oscillate.

3.3 Stability as coefficients vary

Present the Lin panel of the master stability map, with c and received restraint d varied and a=1.5,b=0.65,g=h=0.3 fixed. State the domain and 90 x 90 grid. Explain the gray no-positive-balance region and the numerical stability contour. A compact sentence should explicitly acknowledge stable Lin settings, for example the b=2 counterexample.

End with the limited boundedness observation: finite-time blow-up can occur for sufficiently strong positive feedback, as shown analytically on the synchronous subspace when M<0. This is a conditional limitation of the model, not a statement about the all-ones baseline. The distinction prepares the reader for the modified model.

3.4 Motif dynamics: a separate piece of unfinished work

The record provides candidate two-node equations and proposed analyses, but does not establish a completed comparable motif-results section for both models. Do not convert those suggestions into past-tense results. If your mentor wants the subsection immediately, give the exact reduction and explicitly label the analysis as pending; otherwise reserve the subsection for the next iteration.

Decide whether the remaining nodes are deleted or clamped to fixed background activity: those are different experiments. For a clamped pair, state every background value, derive its two-dimensional system, and later compare its equilibria, phase portrait and response to the corresponding pair in the five-node network. A clamped pair is generally not an invariant isolated subsystem of the full system.

Section 3 figure placement

Use the original-model part of the baseline figures in 3.2 and the original-model map in 3.3. If paired comparison figures are retained, introduce the Lin panels here and return to the modified panels in Section 4. Do not insert duplicate figures merely to make the sections look symmetric.

Plan for Section 4: The modified model and its dynamics

4.1 Specify the model being analyzed

Use the simple five-parameter modified model as the main equation, because it is the model used for the stability graphs. Explain recipient-dependent nourishment, the target-dependent loss and bilinear received restraint. Acknowledge the omitted CHILD term and changed target-effect sign. State the chosen baseline (a,k,c,g,h)=(2,0.5,1,0.5,0.3), and say the values are illustrative rather than fitted.

For nonnegative initial activities and coefficients, this equation preserves nonnegativity. In the current theoretical project it is exactly DM_(5,2). The supplied canonical_model.py contains this equation alone. Lin remains the literature comparator for Section 3; the new project's five-term MB comparator requires its own future comparison figures.

4.2 Positive equilibria and local stability

Write the quadratic (k+h)s^2 + (k+h-a)s - (a+c-g)=0 for nonzero synchronous equilibria. Select and name the upper positive branch used in the maps. State that a+c>g is sufficient for a unique positive root; it is not necessary for the existence of positive roots. The origin always exists.

At the chosen baseline, give x*=(3+sqrt(59))/4 and alpha=-1.153724. Refer to the modified-model eigenvalues. The full circulant derivation can be placed in an appendix or deferred to the theoretical paper. Include the short nonnegativity/boundedness argument if it supports the motivation; clearly distinguish it from asymptotic stability.

4.3 Recovery and parameter dependence

Describe exactly the same relative Wood perturbations as in Section 3, and report recovery to the original balance in these examples. Use a short time window or an inset so the decay can actually be seen. Explain that the five-node symmetry makes perturbing each component in turn redundant at symmetric parameters.

Then discuss the modified stability map using c and received restraint h. In the master window all grid points are locally stable, but the wider historical window includes unstable equilibria. This reconciles the two figures: 100% and about 72% describe different windows and different fixed coefficients, not contradictory calculations.

4.4 Comparison under specified parameter distributions

Briefly describe uniform, ordered and tiered sampling. Use one table with stable counts, valid counts and total draws; a plot is optional. State that the percentages are sampling sensitivity results. Wilson intervals, if shown, quantify Monte Carlo uncertainty conditional on those sampling assumptions, not biological uncertainty.

Prior	Lin: stable / valid / total	Modified: stable / valid / total
Uniform	273 / 1195 / 1200	968 / 1029 / 1200
Ordered	562 / 1192 / 1200	1103 / 1200 / 1200
Tiered	267 / 1145 / 1200	1188 / 1200 / 1200

4.5 Motifs or extensions, if needed for this version

Mirror the explicit reduction chosen in 3.4 and compare like experiments. If those studies are unfinished, say so and end the present section with the established stability comparison. The advanced Hopf and bistability figures are optional supporting material, not prerequisites for this week's write-up.

Reproducibility and drafting checklist

What was independently recomputed

Both Jacobians were checked by complex-step differentiation at 1,000 random states each. Maximum absolute discrepancies were 7.11e-15 (Lin) and 5.33e-15 (modified). Both equilibrium formulas and Fourier real-part formulas were checked over the same parameter draws. The four disturbance runs were compared between DOP853 and RK45.

Both historical stability windows were recomputed on their original 90 x 90 grids. The master Monte Carlo experiments used the original seed and draw order. The CHILD ablation used 3,000 draws and seed 11. The conservative and periodic Lin spectra were independently recomputed by variational integration with QR renormalization every 0.5 time units and a 100-unit transient.

Numerical cautions that should survive editing

The stability maps are finite grids, and a plotted zero contour is interpolated. The expression nanmean(M<0) counts NaNs as false after comparison; it does not omit them. That is why percentages must specify all grid points versus valid equilibria. Near zero eigenvalues, use a numerical tolerance and label the point as marginal/unresolved rather than deciding nonlinear stability from roundoff.

The record's closed-form verification section contains a failed run (missing SymPy) followed by claimed success statistics. This review independently checks the relevant formulas, but that historical traceback is not evidence of a successful rerun. Computational correctness also does not establish an unbiased scientific comparison.

A practical writing order

1. Freeze the simple modified equation and the coefficient notation.
2. Choose the four main figures, and finalize their parameters and captions.
3. Draft 3.1-3.3 and 4.1-4.4 around the verified results.
4. Add a short numerical-methods paragraph and the sampling-count table.
5. Mark motif analysis explicitly as completed or pending.
6. Move regime classification and advanced bifurcations out of the central argument until repaired.

Suggested numerical-methods content

State the ODE solver and tolerances actually used for each submitted figure, the integration interval and initial states, the equilibrium branch, the parameter grid, the sampling distributions and seed, and the stability decision rule. The regenerated disturbance figures use SciPy DOP853 with relative tolerance 1e-10 and absolute tolerance 1e-12. Their sampled time grid has 3,001 points over [0,120].

Source and scope record

Primary supplied sources: *WuXing_Complete_Chat_Record.md*, especially Parts 5-10 and the P4-P5 calculations; and the pasted LaTeX manuscript containing the actual Sections 3 and 4. The local *WuXing_Commented.ipynb* contains no embedded figure outputs. Historical instructions inside these documents were treated as source material, not new user requests.

Original reference checked: Haipeng Lin and Jing Han, *Analysis of Dynamic Five-Element Model*, Proceedings of the 39th Chinese Control Conference (2020), pp. 777-781. DOI: 10.23919/CCC50068.2020.9188438. The local Lin2020.pdf was used to check Eq. (2), baseline eigenvalues, parameter cases and reported figures. This review does not independently validate the historical or clinical claims in the surrounding TCM narrative.

Companion files: the verified-figures folder contains PNG and SVG exports. The revised archive contains the nonnegative-model comparison audit, numerical results and figure exports, plus canonical_model.py and a matching XPPAUT file. The standalone code appendix below contains only the five-term modified model. It does not alter the supplied manuscript or historical notebooks.

Code appendix: canonical five-term model

This is the complete standalone canonical_model.py supplied with the revised package. Its parameter order is (a,k,c,g,h). Lin comparison code is kept separately in audit.py. The exact canonical ODE preserves nonnegative states for the stated parameter domain.

```
"""Five-term demand-modulated model DM_(5,2).

Order: Wood, Fire, Earth, Metal, Water; indices wrap modulo five.
Parameter order: (a, k, c, g, h). Use a,k,c,g,h > 0 and x(0) >= 0.
The exact ODE preserves nonnegativity; numerical solvers have finite error.
This module contains only the canonical model. Requires NumPy.
"""
import numpy as np

BASELINE = (2.0, 0.5, 1.0, 0.5, 0.3)


def rhs(t, x, a, k, c, g, h):
    """Compatible with solve_ivp(..., args=BASELINE)."""
    x = np.asarray(x)
    return (a*x - k*x*x + c*np.roll(x, 1)/(1+x)
            - g*x/(1+np.roll(x, -2)) - h*x*np.roll(x, 2))


def positive_balances(a, k, c, g, h):
    """All positive synchronous levels, sorted; origin exists separately."""
    A = k + h
    D = (A-a)**2 + 4*A*(a+c-g)
    if D < 0:
        return []
    roots = [(a-A-np.sqrt(D))/(2*A), (a-A+np.sqrt(D))/(2*A)]
    return sorted(set(float(s) for s in roots if s > 0))


def jacobian(x, a, k, c, g, h):
    """Jacobian at a five-component state, not only at an equilibrium."""
    x = np.asarray(x)
    J = np.zeros((5, 5))
    for i in range(5):
        mother, target, controller = (i-1) % 5, (i+2) % 5, (i-2) % 5
        J[i, i] = (a - 2*k*x[i] - c*x[mother]/(1+x[i])**2
                   - g/(1+x[target]) - h*x[controller])
        J[i, mother] += c/(1+x[i])
        J[i, target] += g*x[i]/(1+x[target])**2
        J[i, controller] -= h*x[i]
    return J


if __name__ == '__main__':
    s = positive_balances(*BASELINE)[-1]
    alpha = np.linalg.eigvals(jacobian(np.full(5, s), *BASELINE)).real.max()
    print(f'Positive balanced level: {s:.12f}')
    print(f'Largest eigenvalue real part: {alpha:.12f}')
```

Running this file prints x*=2.670286436967 and max Re(lambda)=-1.153723520690. To simulate, pass rhs to scipy.integrate.solve_ivp with args=BASELINE and a nonnegative five-component initial state. A matching canonical_model.ode is included for XPPAUT.